Inertia and balanced forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict motion when the net force is zero.

SUCCESS CRITERIA - I can...

1. Add forces to decide whether they balance.
2. Predict constant velocity when net force is zero.
3. Explain why changing direction requires net force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “inertia” to explain your model or result. Sentence starters are optional.

net force: the vector sum of all forces on one object

inertia: the tendency to keep the same velocity without net force

balanced: adding to zero net force in this force context

01 Warm-up: bring back an earlier idea

A=(6, -4) m and B=(-4, 8) m.
Find R, magnitude and direction.
Choose C so A+B+C=0 and sketch the route.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: ΣF = 0 ⇒ a = 0 ⇒ v = constant.
ΣF means net force. Balanced forces allow rest or steady motion in a straight line.

Worked reasoning: A book has 6 N upward support and 6 N downward weight. Net force is zero. Its velocity stays constant; it could be resting or moving steadily.

Your turn: Draw the two force arrows. Then sketch two possible velocity-time graphs consistent with this balance.

Sentence starter: The … in my model represents … . Its label shows … .



Book: compare support and weight.

Marzano check - circle: ① ② ③

03 Infer the motion constraint

A 4 kg cart experiences 6 N right and 6 N left. It currently moves at -3 m/s.
Find acceleration and its displacement during the next 5 s.

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. Velocity: 3 m/s left

Marzano check - circle: ① ② ③

04 Inertia in a braking bus

Explain a passenger’s forward motion relative to a braking bus using an approximately inertial ground frame.
Identify the external interaction that changes passenger velocity.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

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Continue down each column.

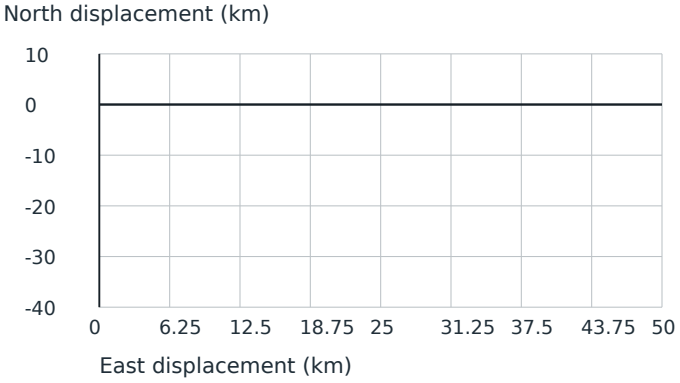
Sentence starters are optional. You may plan in any language.

05 Engine-off spacecraft

A spacecraft coasts in an ideal region with zero net force. Its velocity is (4, -3) km/s, with east +x and north +y.
Draw the displacement over 10 s, find its magnitude, and explain why an engine is not needed to maintain this motion.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A car starts at +15 m/s and brakes at -3 m/s² until rest. A student extends that acceleration to 7 s.
Find the stop time/distance and explain why extension is invalid for ordinary braking.
Sketch the velocity-time model; label the given values.

Word help: acceleration = rate of change of velocity

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

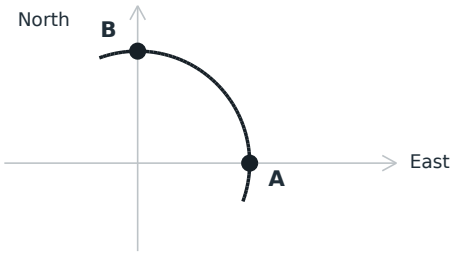


Marzano check - circle: ① ② ③

07 Exit check

A runner goes from A toward B around the shown curve at constant speed. A classmate claims this is equilibrium because speed does not change.
Draw velocity arrows at both points and use them to correct the claim.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

An elevator moves upward at a steady 2 m/s. Its mass is 500 kg; use g = 10 N/kg. Complete its force diagram and find cable tension. Then sketch a case with lower tension while it still moves upward.
Describe how its speed changes.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add and label force arrows.

Choose the elevator as the object.

Marzano check - circle: ① ② ③